

Profile: Senior AI Engineer bridging the gap between ML, Software Engineering, and physical lab environments. 5+ years of experience designing scalable, cloud-based AI systems to accelerate scientific discovery. Expert in Agentic systems that interface with self-driving labs, in-silico data pipelines, and predictive models (GNNs, PINNs) to solve complex search problems in the molecular sciences.

Skills: Python, Rust, TypeScript, PyTorch, Tensorflow, AWS, GCP, Docker, SQL, REST, websockets, LLM-assisted Engineering, French, Spanish.

Senior AI Engineer (Nov 2025 - Present) | AI Engineer (Apr 2024 - Nov 2025) — [Materiom](#)

Funded by Google, leading the technical development and implementation of Materiom's Agentic natural-polymer discovery stack.

- **[Materiom AI](#):** Architected an LLM-powered Multi-Agent framework with tool-use capabilities, RAG over literature/search APIs, and dynamic workflow planning across 100k+ materials, serving 10k+ users and 10+ enterprise partners, with 99.98% uptime.
- **Co-developed evaluation pipelines with cross-functional Chemists and Materials Scientists**, integrating expert domain annotations into feedback loops for targeted agent planning and retrieval diagnostics.
- **ML models:** GNNs, PINNs, and tabular models for materials property prediction & inverse design from experimental & in-silico datasets. Achieving a ~10% error in unseen materials.
- **Closed-loop discovery:** development of Active Learning & Bayesian Optimization workflows for model development and materials discovery. Achieving at least 50% sample efficiency over conventional methods.
- **Self-driving Lab:** design, development & orchestration of Materiom's robotics platform, reaching throughput of 10s of characterized materials per day, including molecular spectroscopy, as well as end material properties.
- **In-silico & experimental data pipelines:** Scaled in-silico data assets 500x by building literature-grounded and CG-MD simulation pipelines using Active Learning to generate high-efficiency training samples.

AI Engineer (Sep 2021 - Apr 2024) — [dRISK](#) (absorbed by [Nexar Inc.](#))

Led the design, development, and deployment of dRISK's Autonomous Vehicle testing, training & validation platform:

- **Scenario-based adversarial validation platform:** developed pipelines for validating RL systems, LiDAR, and camera based perception stacks, specifically validating PPO, CNN & transformer-based architectures under high-risk scenarios.
- Led technical delivery of customer-facing projects for major clients, including **Wayve**, **NVIDIA**, **Luminar**, and **TATA Motors**.

AI Researcher / Master's Thesis (Jun 2021 - Sep 2021) — [Neurolabs](#) & The University of Edinburgh

Thesis: *Graph Convolutional Neural Networks for Synthetic Data Generation using Unity* (Awarded Distinction). Researched and developed geometric deep learning pipelines to generate high-fidelity synthetic spatial data. Read more [here](#).

Education & Scholarships

MSc in Artificial Intelligence ([The University of Edinburgh](#), grade: distinction), MSc in Mechatronic Engineering ([Arts et Métiers ParisTech](#), grade: 17/20), BSc in Product Design Engineering ([EAFIT University](#), grade: 4.75/5.0). [Colfuturo Excellence Scholarship](#), [Eiffel Excellence Scholarship](#).

Patents & Applied Research

- **Material discoveries undergoing patentability reviews:** Engineered the AI discovery stack at Materiom that produced 5+ breakthrough natural-polymer candidates currently undergoing patenting and commercial trialing.
- ([Application No. PCT/US2024/032276](#)): ML algorithm using graph theory to augment trajectory data for training AV RL models.

Talks

AI for Materials Science, University of the Arts Central Saint Martins, [A day in the life of an AI Startup](#), EAFIT University, [Synthetic Data Generation for ML](#), Machine Learning Colombia